Delta Hedging in Discrete Time:

Rebalancing Error, Volatility Misspecification and Transaction Costs

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October 2026

Abstract

In the Black–Scholes model a European call can be replicated by trading in the underlying stock, provided the hedge is adjusted continuously, without costs, and with the right volatility. This note studies, by Monte Carlo simulation, what happens when each of these conditions fails: the hedge is rebalanced at finitely many dates, the volatility used to hedge differs from the one the stock actually exhibits, and every trade incurs a proportional cost. A single second-order Taylor expansion of the option price over one rebalancing period explains all three effects. The rebalancing error decays like $N^{-1/2}$ in the number N of rebalancing dates; a misspecified volatility produces a profit equal to a gamma-weighted gap between the two variances, which I verify path by path; and transaction costs create an expected loss growing like $N^{1/2}$, which Leland's adjusted volatility, interpreted as a deliberate volatility misspecification, offsets almost entirely. All code is available at `https://github.com/USERNAME/delta-hedging`.

1 Introduction

The Black–Scholes price of an option is the initial cost of a self-financing portfolio that replicates its payoff [2]. Replication requires trading continuously, in a frictionless market, with knowledge of the true volatility. None of these holds in practice, so a trader who sells an option and hedges it is exposed to a residual profit and loss (P&L).

This note quantifies that residual in a controlled setting: the stock follows the Black–Scholes model, and a trader sells a three-month at-the-money call and delta-hedges it. Three questions are studied:

1. How large is the hedging error when the hedge is rebalanced only N times? (Section 3)
2. What happens when the trader hedges with a volatility different from the one the stock exhibits? (Section 4)
3. How do proportional transaction costs change the picture, and can they be priced in? (Section 5)

The answers are classical [6, 4, 5, 1, 3]. The aim is to show that one elementary computation, a second-order Taylor expansion over a single rebalancing period, explains all three, and to check each prediction numerically.

Level of rigour. The derivations are heuristic: they use Taylor expansions, keep only leading-order terms, and rely on elementary properties of Gaussian random variables (collected in Appendix A). Rigorous versions of these results are formulated in continuous time and require stochastic calculus; references are given where relevant. The simulations, by contrast, involve no approximation other than the discreteness of the hedge.

2 Setting

Model. Time runs from 0 to the maturity T and is divided into N periods of length $\Delta t = T/N$, with dates $t_i = i\,\Delta t$. Over each period the stock price is multiplied by an independent lognormal factor:

$$S_{t_{i+1}} = S_{t_i} \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_i \Big), \qquad Z_0, Z_1, \dots \text{ i.i.d. } \mathcal{N}(0,1). \tag{1}$$

Since $\mathbb{E}[e^{\sigma\sqrt{\Delta t}Z}] = e^{\sigma^2\Delta t/2}$, the expected return over a period is $\mathbb{E}[S_{t_{i+1}}/S_{t_i}] = e^{\mu\Delta t}$: μ is the drift. The log-return has variance $\sigma^2\Delta t$: σ is the volatility. This is exactly the Black–Scholes model (geometric Brownian motion) observed at the dates t_i; in particular, the simulations below are exact. Cash earns the constant rate r.

Black–Scholes formulas. The Black–Scholes price of a call with strike K is

$$C(t,S) = S\,\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r \pm \tfrac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad \tau = T - t,$$

where Φ and φ are the standard normal distribution function and density. Its sensitivities are the delta $\Delta = \partial_S C = \Phi(d_1)$, the gamma $\Gamma = \partial_{SS} C = \varphi(d_1)/(S\sigma\sqrt{\tau}) > 0$ and the vega $\mathcal{V} = \partial_\sigma C = S\varphi(d_1)\sqrt{\tau}$. A direct computation (Appendix B) shows that C satisfies the Black–Scholes equation

$$\partial_t C + \tfrac{1}{2}\sigma^2 S^2\,\Gamma + rS\,\Delta - rC = 0. \tag{2}$$

Throughout, $S_0 = K = 100$, $T = 0.25$, $r = 2\%$, $\mu = 8\%$ and $\sigma = 20\%$, so that $C(0,S_0) = 4.23$.

Hedging strategy. The trader sells the call at time 0, buys Δ_0 shares and keeps the rest in cash. At each date t_i, $i = 1, \dots, N-1$, the position is adjusted to Δ_{t_i} shares, the trade being financed from cash; no money is added or withdrawn, so the strategy is self-financing. At maturity the trader pays $(S_T - K)^+$. The terminal P&L is

$$\text{P\&L} = B_T + \Delta_{t_{N-1}} S_T - (S_T - K)^+,$$

where B_T is the cash account. The Black–Scholes replication result says that P&L $\to 0$ as $N \to \infty$; Section 3 quantifies how fast.

Simulation. Paths are generated with (1). Results use 20,000 paths for the main figures and 10,000 paths per point in the parameter sweeps.

3 Discrete rebalancing

Local error. Consider one period $[t_i, t_{i+1}]$ and write $\delta S = S_{t_{i+1}} - S_{t_i}$. The trader is short one option, long Δ shares, and holds $C - \Delta S$ in cash. A second-order Taylor expansion of C gives, to leading order,

$$\delta C \approx \partial_t C\,\Delta t + \Delta\,\delta S + \tfrac{1}{2}\Gamma\,\delta S^2,$$

while the shares gain $\Delta\,\delta S$ and the cash earns $r(C - \Delta S)\Delta t$. The change in the value of the position is therefore

$$-\delta C + \Delta\,\delta S + r(C - \Delta S)\Delta t \approx -\partial_t C\,\Delta t - \tfrac{1}{2}\Gamma\,\delta S^2 + r(C - \Delta S)\Delta t.$$

Replacing $\partial_t C$ by its value from (2), all terms in C and Δ cancel and only a convexity term remains:

$$\varepsilon_i \approx \tfrac{1}{2}\,\Gamma_{t_i} S_{t_i}^2 \left(\sigma^2\Delta t - \left(\frac{\delta S}{S_{t_i}}\right)^2 \right). \tag{3}$$

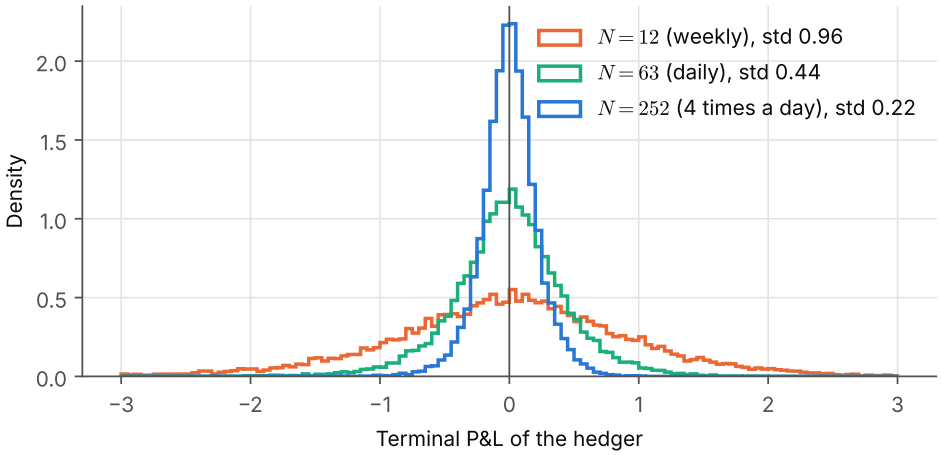


Figure 1: Distribution of the terminal P&L of a delta-hedged short call for three rebalancing frequencies. The option is worth 4.23.

By (1), $\delta S/S_{t_i} = e^x - 1$ with $x = (\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_i$, so $(\delta S/S_{t_i})^2 = \sigma^2 \Delta t\, Z_i^2$ up to terms of order $\Delta t^{3/2}$. Hence

$$\varepsilon_i \approx a_i \left(1 - Z_i^2\right), \qquad a_i = \tfrac{1}{2}\,\Gamma_{t_i} S_{t_i}^2\, \sigma^2 \Delta t. \tag{4}$$

Two remarks. First, the terms in $\Delta\,\delta S$ cancelled exactly, so the drift μ, which only enters through δS, plays no role at this order. Second, a_i depends only on $Z_0, \dots, Z_{i-1}$, while Z_i is independent of them. Since $\mathbb{E}[Z_i^2] = 1$, each local error has mean zero.

Global error. The local errors are uncorrelated. Indeed, for $i < j$, the product $\varepsilon_i a_j$ depends only on $Z_0, \dots, Z_{j-1}$ and is therefore independent of Z_j, so

$$\mathbb{E}[\varepsilon_i \varepsilon_j] = \mathbb{E}[\varepsilon_i a_j]\; \mathbb{E}[1 - Z_j^2] = 0.$$

By the same independence and $\mathrm{Var}(Z^2) = 2$ (Appendix A), $\mathbb{E}[\varepsilon_i^2] = 2\,\mathbb{E}[a_i^2]$. The variance of the total error is then the sum of the variances, which is a Riemann sum:

$$\mathrm{Var}(\text{P\&L}) \approx 2\sum_i \mathbb{E}[a_i^2] = \frac{\Delta t}{2}\sum_i \mathbb{E}\left[(\Gamma_{t_i} S_{t_i}^2 \sigma^2)^2\right]\Delta t \approx \frac{T}{2N}\int_0^T \mathbb{E}\left[(\Gamma_t S_t^2 \sigma^2)^2\right]\mathrm{d}t.$$

The standard deviation of the P&L therefore decays like $N^{-1/2}$. Evaluating the integral for an at-the-money option, Derman and Kamal [4] obtain the explicit approximation

$$\mathrm{Std}(\text{P\&L}) \approx \sqrt{\frac{\pi}{4}}\,\frac{\sigma\,\mathcal{V}}{\sqrt{N}}, \qquad \mathcal{V} = \partial_\sigma C(0, S_0). \tag{5}$$

Results. Figure 1 shows the P&L distribution for weekly ($N = 12$), daily ($N = 63$) and four-times-daily ($N = 252$) rebalancing. The distributions are centred at zero (means -0.009, -0.001 and $+0.001$) with standard deviations 0.96, 0.44 and 0.22: multiplying N by four roughly halves the error. Figure 2 confirms the rate: the fitted log–log slope is -0.489, against -0.5 in theory, and (5) is within 8% of the simulation over the whole range.

Role of the drift. The cancellation of the drift only holds to leading order. Hedging under drifts of -20% and $+40\%$, the mean P&L moves by at most 0.028 with daily rebalancing and by at most 0.0007 with $N = 1000$: the drift affects the hedge only through higher-order terms, which vanish as $N \to \infty$.

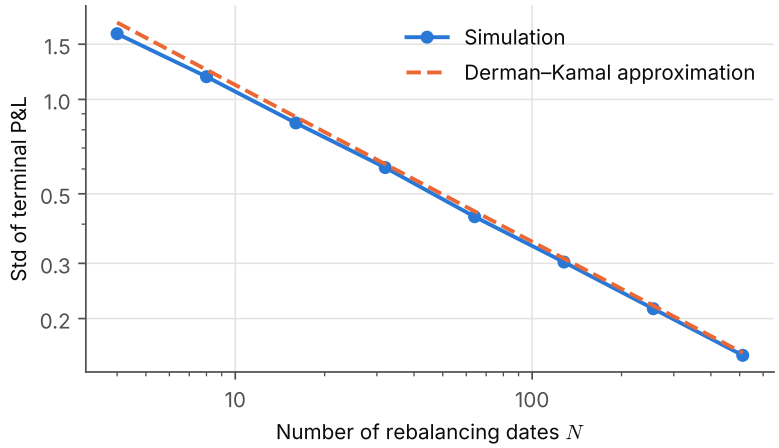


Figure 2: Standard deviation of the P&L against N (log–log scale), with the Derman–Kamal approximation (5).

4 Hedging with the wrong volatility

In practice the volatility is not observed. Options trade at an *implied* volatility Σ, while the stock moves with some volatility σ that is only known after the fact (the *realised* volatility). Here the stock follows (1) with $\sigma = 15\%$, while the option is sold at $\Sigma = 20\%$. Costs are set to zero.

Hedging at the implied volatility. Let C^Σ, Δ^Σ, Γ^Σ denote the Black–Scholes quantities computed with Σ, and suppose the trader sells at C^Σ and hedges with Δ^Σ. The computation of Section 3 goes through unchanged, except that C^Σ satisfies (2) with Σ in place of σ. The local error (3) becomes

$$\varepsilon_i \approx \tfrac{1}{2}\,\Gamma^\Sigma_{t_i} S^2_{t_i} \left(\Sigma^2 \Delta t - \left(\frac{\delta S}{S_{t_i}} \right)^2 \right),$$

and, using again $(\delta S / S_{t_i})^2 \approx \sigma^2 \Delta t\, Z_i^2$, it splits into two parts:

$$\varepsilon_i \approx \underbrace{\tfrac{1}{2}\,\Gamma^\Sigma_{t_i} S^2_{t_i} \left(\Sigma^2 - \sigma^2 \right) \Delta t}_{\text{systematic gain}} + \underbrace{\tfrac{1}{2}\,\Gamma^\Sigma_{t_i} S^2_{t_i}\, \sigma^2 \Delta t \left(1 - Z_i^2 \right)}_{\text{rebalancing noise}}. \tag{6}$$

The second part has exactly the structure of (4): by the argument of Section 3, its total has standard deviation of order $N^{-1/2}$ and vanishes as $N \to \infty$. The first part is a gain realised at time t_i; it stays in the cash account until maturity, where it has grown by the factor $e^{r(T-t_i)}$. Summing over i gives a Riemann sum, and in the limit

$$\text{P\&L} = \frac{\Sigma^2 - \sigma^2}{2} \int_0^T e^{r(T-t)}\, \Gamma^\Sigma_t S^2_t \, \mathrm{d}t. \tag{7}$$

A rigorous continuous-time version of this formula is due to El Karoui, Jeanblanc-Picqué and Shreve [5]. Since $\Gamma > 0$, the seller makes money on *every* path when the implied volatility exceeds the realised one. The amount, however, depends on the path: it is large when the stock stays close to the strike near maturity, where gamma is concentrated.

Hedging at the realised volatility. Alternatively, the trader sells at C^Σ but hedges with Δ^σ. By Section 3, this hedge replicates a call priced at C^σ as $N \to \infty$, so the trader keeps the price difference, compounded to maturity:

$$\text{P\&L} \to e^{rT} \left(C^\Sigma(0, S_0) - C^\sigma(0, S_0) \right) = 0.997. \tag{8}$$

The profit is deterministic, but the strategy requires knowing σ in advance.

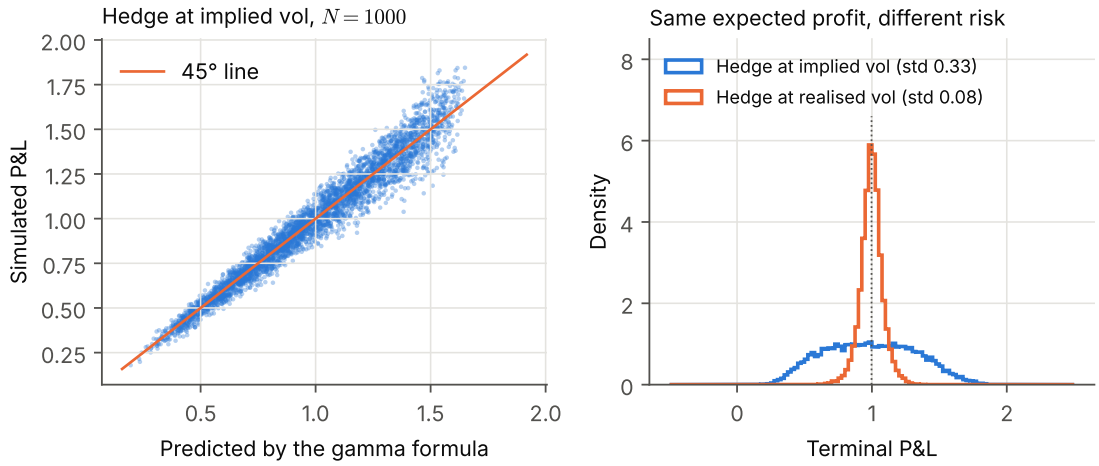


Figure 3: Option sold at implied volatility $\Sigma = 20\%$, stock with realised volatility $\sigma = 15\%$. Left: simulated P&L of the implied-volatility hedge against formula (7), on 4,000 paths. Right: P&L of the implied-volatility and realised-volatility hedges; the dotted line is the locked-in profit (8).

	Hedge at implied Σ		Hedge at realised σ	
N	mean	std	mean	std
16	0.986	0.703	0.992	0.627
64	0.987	0.447	0.994	0.318
256	0.987	0.362	0.999	0.164
1024	0.995	0.335	0.997	0.083

Table 1: P&L of the two hedging strategies when $\Sigma = 20\%$ and $\sigma = 15\%$ (10,000 paths per row). Only the realised-volatility hedge converges to a deterministic profit.

Results. The left panel of Figure 3 plots, path by path, the simulated P&L of the implied-volatility hedge against the prediction (7), evaluated on the same path. With $N = 1000$ the correlation is 0.978, and the residual, which is the rebalancing noise in (6), has standard deviation 0.070. The P&L is positive on all 20,000 paths, ranging from 0.16 to 1.92.

The right panel compares the two strategies. Both earn about the price gap on average (0.988 and 0.997, against 0.997 from (8)), but their risks differ: 0.334 for the implied hedge against 0.083 for the realised one. Table 1 shows that the second number tends to zero as N grows, while the first levels off: its randomness is not a discretisation effect but is intrinsic to (7).

5 Transaction costs

Cost model. We return to the setting of Section 3, where the true volatility σ is known, and add trading costs: each trade of n shares at price S costs $\frac{k}{2}|n|S$, where k is the round-trip proportional cost. I take $k = 0.4\%$. All trades, including the initial purchase of Δ_0 shares, are charged.

Scaling of the cost. Over one period the hedge ratio changes, to leading order, by $\delta\Delta \approx \Gamma\,\delta S \approx \Gamma S\sigma\sqrt{\Delta t}\,Z$. Since $\mathbb{E}|Z| = \sqrt{2/\pi}$ (Appendix A), the expected cost of one rebalance is

$$\mathbb{E}[\text{cost}_i] \approx \frac{k}{2}\sqrt{\frac{2}{\pi}}\,\sigma\,\Gamma_{t_i}S_{t_i}^2\sqrt{\Delta t}. \tag{9}$$

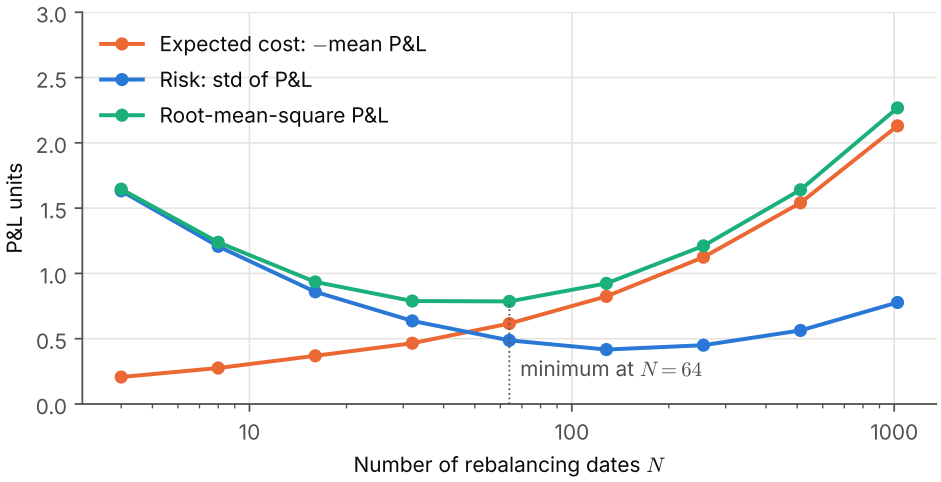


Figure 4: Hedging with transaction costs ($k = 0.4\%$ round trip): expected cost, risk, and root-mean-square P&L against the number of rebalancing dates.

The total expected cost, a sum of N such terms, grows like $N\sqrt{\Delta t} = \sqrt{NT}$. Rebalancing more often reduces the error of Section 3, which shrinks like $1/\sqrt{N}$, but the bill grows like $\sqrt{N}$.

Results. Figure 4 shows the trade-off. The expected loss increases steadily with N and reaches 2.13, half the option's value, at $N = 1024$. The standard deviation is no longer monotone: it falls to 0.42 at $N = 128$ and then rises again, because the random amount of costs paid becomes itself a source of risk. The root-mean-square P&L, which penalises both bias and risk, is minimised at $N = 64$, roughly daily rebalancing. Exact replication is no longer possible, and selling at the Black–Scholes price is a losing strategy on average.

Leland's correction. Section 4 suggests a remedy. By (6), a trader who prices and hedges with a volatility $\hat{\sigma}$ larger than the true σ earns, at each step, the systematic gain $\frac{1}{2}\Gamma S^2(\hat{\sigma}^2 - \sigma^2)\Delta t$. Like the expected cost (9), this gain is proportional to ΓS^2. Choosing $\hat{\sigma}$ so that the two are equal at every step,

$$\tfrac{1}{2}\,\Gamma S^2\big(\hat{\sigma}^2 - \sigma^2\big)\Delta t = \frac{k}{2}\sqrt{\frac{2}{\pi}}\,\sigma\,\Gamma S^2\sqrt{\Delta t},$$

gives Leland's volatility [6]:

$$\hat{\sigma}^2 = \sigma^2\left(1 + \sqrt{\frac{2}{\pi}}\,\frac{k}{\sigma\sqrt{\Delta t}}\right). \tag{10}$$

Leland's correction is thus a deliberate volatility misspecification: the trader sells the option at the Black–Scholes price computed with $\hat{\sigma}$, hedges with the corresponding delta, and the extra premium pays for the costs. For daily rebalancing, $\hat{\sigma} = 22.4\%$ and the option is sold for 4.71 instead of 4.23.

Figure 5 compares the two strategies. With Leland's volatility the mean P&L stays between -0.111 and -0.093 for every N. This residual is the cost of buying the initial hedge, $\frac{k}{2}\Delta_0 S_0 = 0.108$, a one-off trade that (10), built from rebalancing costs, does not price. The risk also keeps decreasing with N, to 0.13 at $N = 1024$ against 0.78 for the Black–Scholes hedge.

The derivation of (10) treats $k/(\sigma\sqrt{\Delta t})$ as a moderate correction. As $\Delta t \to 0$ with k fixed, $\hat{\sigma}$ grows without bound, and the approximation breaks down: Soner, Shreve and Cvitanić [7] showed that in continuous time with proportional costs, the cheapest way to hedge a call with certainty is to buy one share.

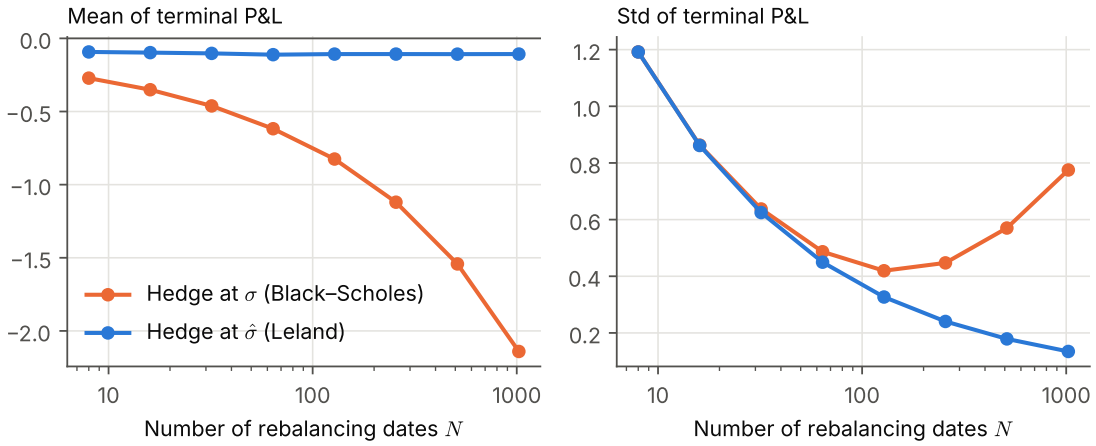


Figure 5: Mean (left) and standard deviation (right) of the P&L under transaction costs, when pricing and hedging with the true volatility σ or with Leland's volatility $\hat{\sigma}$.

6 Conclusion

A single second-order Taylor expansion over one rebalancing period, equation (6), accounts for all three effects studied here. Its random part, of mean zero, gives a rebalancing error of order $N^{-1/2}$. Its systematic part gives the profit of hedging with a volatility different from the realised one, which is the gamma-weighted gap (7). Transaction costs add a loss of order $N^{1/2}$, which Leland's correction offsets by using that systematic part on purpose. In every case the simulations agree with the leading-order predictions.

Two directions seem natural to pursue. First, the arguments here are heuristic and limited to leading order. Making them rigorous, and going beyond Leland's approximation, requires stochastic calculus: with transaction costs, exact pricing leads to nonlinear equations that can be formulated as backward stochastic differential equations (BSDEs) [3]. Learning these tools, and solving such equations numerically to compare with Leland's formula, is the natural continuation of this work. Second, the hedging dates were fixed in advance; rebalancing only when the delta leaves a band around its target is a simple way to trade fewer shares for the same risk, and comparing it with calendar hedging would refine the trade-off of Figure 4.

Reproducibility. Every figure and every number in this note is produced by a single script, `experiments/run_experiments.py`, which writes the numbers into the LaTeX source. The code, with unit tests, is available at `https://github.com/USERNAME/delta-hedging`.

A Gaussian moments

Let $Z \sim \mathcal{N}(0,1)$ with density $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$. Using $\varphi'(z) = -z\varphi(z)$:

- $\mathbb{E}|Z| = 2\int_0^\infty z\,\varphi(z)\,\mathrm{d}z = 2\big[-\varphi(z)\big]_0^\infty = 2\varphi(0) = \sqrt{2/\pi}$.
- Integrating by parts, $\mathbb{E}[Z^{n+2}] = \int z^{n+1}\cdot z\varphi(z)\,\mathrm{d}z = (n+1)\,\mathbb{E}[Z^n]$. Hence $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, and $\mathrm{Var}(Z^2) = 3-1 = 2$.
- $\mathbb{E}[e^{aZ}] = \int e^{az-z^2/2}\,\mathrm{d}z/\sqrt{2\pi} = e^{a^2/2}$, by completing the square.

B The Black–Scholes equation

All partial derivatives of C follow from one identity. Since $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = 2\big(\ln(S/K) + r\tau\big)/(\sigma\sqrt{\tau})$,

$$\frac{\varphi(d_2)}{\varphi(d_1)} = \exp\Big(\frac{d_1^2 - d_2^2}{2}\Big) = \exp\big(\ln(S/K) + r\tau\big) = \frac{Se^{r\tau}}{K}, \qquad \text{that is,} \qquad S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2).$$

Delta and gamma. Since $\partial_S d_1 = \partial_S d_2 = 1/(S\sigma\sqrt{\tau})$, differentiating C gives $\Phi(d_1) + \big(S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\big)\partial_S d_1 = \Phi(d_1)$, by the identity. Differentiating once more, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$.

Vega. Similarly, $\partial_\sigma C = S\varphi(d_1)\,\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$.

Time derivative. Time enters through $\tau = T - t$, so $\partial_t = -\partial_\tau$. Using the identity and $\partial_\tau(d_1 - d_2) = \sigma/(2\sqrt{\tau})$,

$$\partial_t C = -\frac{\sigma S\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2).$$

Conclusion. Substituting, $\frac{1}{2}\sigma^2 S^2\Gamma = \sigma S\varphi(d_1)/(2\sqrt{\tau})$ cancels the first term of $\partial_t C$, and $rS\Delta - rC = rKe^{-r\tau}\Phi(d_2)$ cancels the second, which proves (2).

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